

- 2 A skydiver jumps from an aircraft at time $t = 0$ and falls vertically downwards. The variation with t of her velocity v is shown in Fig. 2.1.

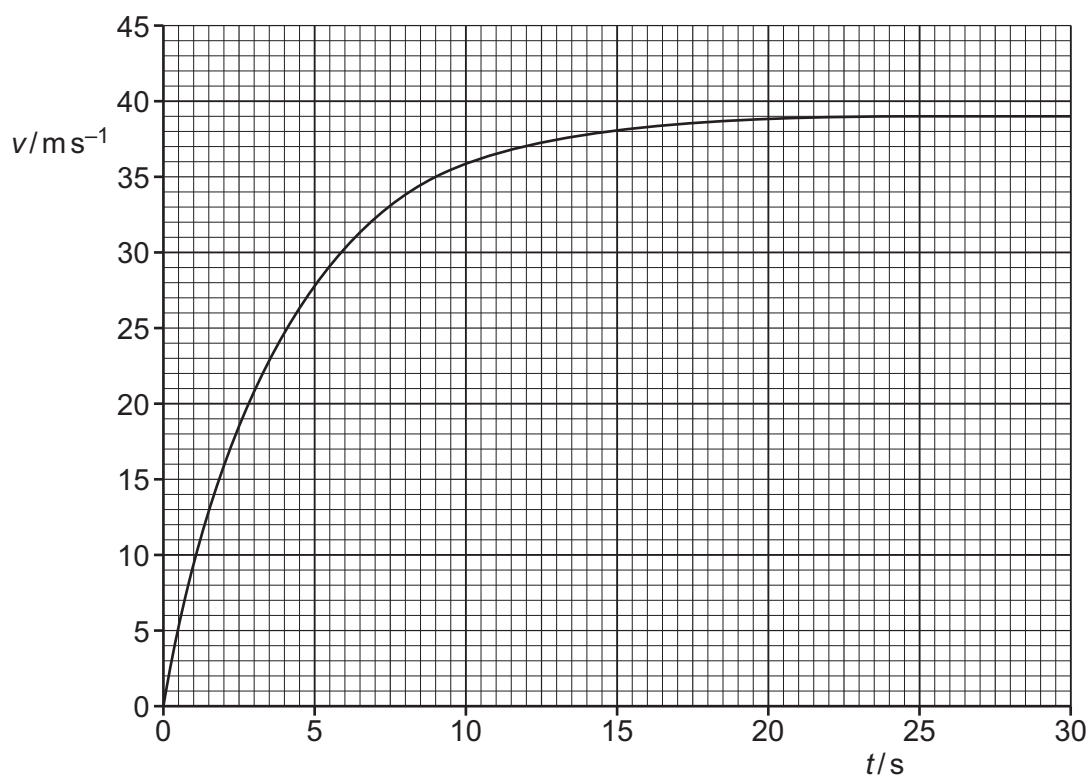


Fig. 2.1

- (a) (i) Using Fig. 2.1, state the terminal velocity of the skydiver.

terminal velocity = ms^{-1} [1]

- (ii) By drawing a suitable line on Fig. 2.1, determine the acceleration of the skydiver at time $t = 9.0 \text{ s}$.

acceleration = ms^{-2} [2]



- (b) The mass of the skydiver and her equipment is 68 kg. The upthrust on the skydiver is negligible.

After reaching terminal velocity, the skydiver opens her parachute at time t_1 . A total drag force of 1800 N acts on the skydiver.

Determine the magnitude and direction of the acceleration of the skydiver at time t_1 .

acceleration = ms^{-2}

direction = [3]

- (c) The parachute is fully open at time t_2 . At a later time t_3 the skydiver reaches a constant velocity of 5.7 ms^{-1} .

- (i) Describe and explain the variation with time of the magnitude of her acceleration between time t_2 and time t_3 .

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 [2]

- (ii) Calculate the change in momentum of the skydiver between time t_1 and time t_3 .

change in momentum = N s [2]

[Total: 10]

